Choosing a batch-correction method for multi-site single-cell studies

⚠️ **Synthetic example.** Every finding, number and citation below is fabricated. No source in the list resolves, and none should be quoted or relied on. This file exists to show the report format — the TL;DR, the evidence table with explicit confidence, the contested section, and recommendations tied back to numbered findings. A real report from this agent cites only sources it actually retrieved.

Date: 2026-09-08 · **Audience:** lab computational staff · **Depth:** Standard

Question

Which batch-correction method should a lab standardize on for single-cell RNA-seq collected across multiple sites — and does the choice change when the biological signal of interest is subtle? Informs a decision about the lab's default pipeline.

TL;DR

- **No method wins outright.** The ranking flips depending on whether batch and biological condition are confounded [1][4].
- For **unconfounded** designs, the three leading methods are within noise of one another; pick on runtime and maintenance, not accuracy [2].
- For **confounded** designs — the common case in multi-site work — aggressive correction removes real biological signal along with batch effect [4][7].
- **Standardize on a default, not a mandate.** The evidence supports one default with a documented escape hatch for confounded designs [1][6].
- The single highest-value change is not the method but **recording batch structure at collection time** [8].

Scope

- **Covered:** integration methods for scRNA-seq, 2019–2025, human and mouse; benchmark studies with ≥3 datasets
- **Excluded:** spatial and multiome integration; methods without a maintained implementation
- **Assumptions:** "multi-site" means ≥2 collection sites with independent library prep; not merely multiple sequencing runs
- **Sources:** 8 total (6 peer-reviewed, 1 preprint, 1 benchmark repository) · window: 2019–2025

Findings

How much does method choice actually matter?

- Across 12 benchmark datasets, the top three methods differed by <4% on batch-mixing metrics — smaller than the spread from parameter choice within a single method [2]
- Runtime differed by ~40× across the same methods, which matters more in practice than the accuracy gap [2]
- *Single-sourced:* one group reports a larger gap on very large datasets [9]

What happens when batch and condition are confounded?

- Correction strength that is appropriate for unconfounded data removed 30–60% of true differential expression when condition was fully confounded with site [4]
- The effect is monotonic in correction strength — there is no setting that removes batch effect without touching confounded biology [4][7]
- No benchmark in this set evaluates partial confounding, which is the realistic middle case [gap — see below]

Does the biological signal's magnitude change the answer?

- For large effects (whole-cell-type shifts), method choice was near-irrelevant [2][3]
- For subtle effects (within-cell-type state changes), methods diverged sharply, and the most aggressive method performed worst [4]

Evidence Summary

Claim	Support	Strongest source	Confidence
Top methods are within noise when unconfounded	2 benchmarks, 12 datasets	[2]	High
Aggressive correction destroys confounded signal	1 benchmark, 1 simulation	[4]	Moderate
Runtime gap exceeds accuracy gap	1 benchmark	[2]	Moderate
Method choice matters more for subtle effects	1 benchmark, 1 reanalysis	[4]	Low

Confidence levels follow `resources/source-quality.md`.

Contested & Uncertain

- **Whether batch-mixing metrics measure the right thing** — one group argues they reward over-correction by construction [7]; the benchmark authors argue paired biological-conservation metrics already control for this [2]. Unresolved because the two camps use different ground-truth definitions.

- **Whether deep-learning methods generalize** — strong in-benchmark results [9] versus reported failures on out-of-distribution tissue [5]. Too few independent replications to call.

Gaps

- **Partial confounding is unevaluated.** Every benchmark here uses either fully confounded or fully crossed designs; real studies sit in between.
- **No study reports cost of being wrong** — i.e. downstream false-discovery consequences of over-correction.
- A search for prospective (rather than retrospective) evaluations returned nothing. Absence worth noting: this literature is entirely reanalysis.

Recommendations & Next Steps

Recommendations

- **Adopt one default method for unconfounded designs, chosen on runtime and maintenance** — because accuracy differences are within noise [2]. Confidence: High.
- **Require an explicit review when batch and condition are confounded** — do not apply the default; the failure mode is silent loss of real signal [4][7]. Confidence: Moderate.
- **Avoid selecting a method on benchmark leaderboards alone** — the metrics themselves are contested [7]. Confidence: Moderate.
- **Do not standardize on a deep-learning method yet** — generalization evidence is thin and contested [5][9]. Confidence: Low; revisit if independent out-of-distribution replications appear.

Next steps

#	Action	Why	Effort
1	Add batch/site fields to the intake sheet	Closes the highest-value gap [8]; makes confounding visible before analysis	Low
2	Run the two candidate defaults on one in-house dataset	Converts a literature call into a local one	Med
3	Write the confounded-design escape hatch into the pipeline docs	Makes recommendation 2 enforceable	Low

Sources

1. Example, A., & Placeholder, B. (2023). *A fabricated review of integration strategies*. Journal of Example Methods. [synthetic — does not exist]
2. Sample, C. et al. (2022). *A fabricated benchmark of twelve datasets*. Journal of Example Methods. [synthetic — does not exist]

3. Demo, D. (2021). *A fabricated reanalysis*. Example Reports. [synthetic — does not exist]
4. Placeholder, B. et al. (2024). *A fabricated study of confounded designs*. Example Genomics. [synthetic — does not exist]
5. Fictional, E., & Demo, D. (2025). *A fabricated out-of-distribution evaluation*. Example Letters. [synthetic — does not exist]
6. Example, A. (2020). *Fabricated pipeline guidance*. Example Practice. [synthetic — does not exist]
7. Sample, C. (2023). *A fabricated critique of mixing metrics*. Example Commentary. [synthetic — does not exist]
8. Demo, D. et al. (2024). *Fabricated recommendations on study design capture*. Example Design. [synthetic — does not exist]
9. Fictional, E. et al. (2024). *A fabricated deep-learning benchmark* [preprint]. Example Preprints. [synthetic — does not exist]